08. THE HEART : DYNAMICS OF DEVELOPMENTAL MOVEMENT

DURATION : 11:04 STUDY SHEET

COURSE SUMMARY

This video delves into the heart of cardiac embryology, exploring the fundamental stages of heart growth in early embryonic development. It addresses key concepts such as the approximation of the endocardial tubes, the 'looping' movement, and the inversion processes that transform the heart. The instructor emphasises the importance of understanding the dynamic between the heart and the liver, as well as the need to follow the embryonic blueprint to better support patients. By integrating this knowledge, practitioners will be able to refine their biodynamic approach, thus promoting harmonious development in their patients.

THE HEART: DYNAMICS OF DEVELOPMENTAL MOVEMENT

INTRODUCTION TO EMBRYONIC CARDIAC DEVELOPMENT

We will observe the **exponential growth** and **coiling** of the embryo.

Here, we only see the **vitelline cavity**. Above it, the **amniotic cavity** has been cut.

There is **longitudinal growth** and **flexion**. We can distinguish the **nervous system**, the **somites**, and the **vascular systems**.

Let's focus on an opening, the **future intraembryonic coelom orifice**. We must imagine it here. It will later transform this external coelom into an internal coelom.

The impressive growth of the system brings the two **primitive endocardial tubes** closer to the midline, due to a **differential growth rate**, influenced by the **notochord**.

This approximation of the endocardial tubes forms, above, the **intestine**, and all around, the **intraembryonic cavity**. At this stage, this will form the **pericardial cavity**, not the peritoneal cavity.

The **vitelline vesicle** moves closer to form the upper part of the intestine.

KEY MOVEMENTS OF CARDIAC DEVELOPMENT

The approximation of the two endocardial tubes forms the area of the developing **cardiac tube**.

We observe a movement of approximation towards the midline and, simultaneously, a movement where the brain "does this". It's a combination of both.

This process is synchronous with the **flexion** phase of the embryo. The meeting of the thumbs symbolises this first movement at the level of the heart.

THE HEART'S "LOOPING" MOVEMENT

The heart, in sagittal view, meets and seeks to develop. It lacks space behind it, and little space to the left or right. It only has space forwards.

Within the heart itself, it will find its own **fulcrum**. With the back blocked, it develops forwards, performing what is called the heart's **"looping" movement**. This is phase 2.

This continuous development forms a third level. The space between the **ventricular system** and the **atrium** becomes a fulcrum. The ventricle, initially above, ends up below in relation to the system. This is an **inversion** movement.

Here are the three main movements:

1. **Approximation** towards the midline and meeting.

2. **"Looping" movement** forwards.

3. **Inversion process** of the ventricle.

THE INFLUENCE OF THE PERITONEAL CAVITY AND SYMMETRY BREAKING

The third movement is linked to the inversion process, which is a developmental process of the **peritoneal cavity**.

The development of the **liver** causes a shift in movement. The lesser sac displaces the left bronchus higher than the right, linked to the **breaking of axial symmetry**.

We therefore observe a developmental movement forwards, but also to the left. This is a **double tilt**.

Let's mainly remember:

- The heart's **approximation** towards the midline.
- **Development** forwards.
- Simultaneous development to the left.

The **sinusoid system** of the liver will position itself upwards after this tilt.

ORGANISATION OF THE GREAT VESSELS

The first movement, then the second movement linked to the tilt, and finally a third movement that organises the **great vessels** above.

The heart is tilted, held by an axis, and balanced in relation to the **vascular plane**. There is greater development on one side.

It is a balance between the **thoracic** part, the **abdominal** part, and the **lower visceral** part. A very important movement occurs between the heart, stomach, and liver.

THE EMBRYONIC IMPRINT IN A BIODYNAMIC APPROACH

A very important reference point to rebalance is the **hepatic movement**. We observe whether the liver is well within its developmental movement or if it is receding.

If the liver stops, there is a small halt.

At the level of the heart, its development will do the opposite. There is a trajectory, like an **imprint**.

One must position oneself on the embryonic imprint, go where the system wishes to go; that is where the **stillness** is.

If the movement does not go far enough, the system must be helped. This organisation of the heart is strongly influenced by the liver. Hence the importance of understanding the role of the liver.

In biodynamic embryology, we do not impose. We reject neither **fascial restrictions**, nor the **fluid** aspect, nor **structural**, **visceral** or other techniques.

The approach is to adapt to what is felt in the present moment. The therapeutic force lies in the meeting of the two neutrals.

The regenerative processes in **biodynamic osteopathy** use these embryonic movements. It is not about direct therapy, but about recognising the embryonic developmental movements that appear in therapeutic processes guided by the **"tide"**.

It is crucial to recognise the frequency one is on.

THE HEART'S ENERGY AND ITS ENVIRONMENT

We rarely touch the heart directly first. The heart forms through the **echo** of the periphery.

The heart is extremely powerful because it receives very important information from the **neural crest**. It receives even more powerful electrical information on a neurological level.

This means that one can learn to feel the heart at a distance, or that the heart is felt at a distance. It has a power, a **radiation**. Measurements have shown radiation up to at least 5 metres from the person, probably linked to the neural crest.

It is on the **venous** plane that there will be development, called the **venous sinus** of the heart. In this venous sinus, the **cardinal veins**, **umbilical veins**, and **vitelline veins** will converge, undergoing many rearrangements.

The heart has a **fascial** environment and its own **electricity** for its beat. However, this electricity was not sufficient to maintain the heart in its proper function. It depends heavily on its entire fascial environment.

The **sterno-pericardial**, **phrenico-pericardial**, **vertebro-pericardial ligaments**, and the **superior aorta**, suspended up to the pharyngeal tubercle, support the heart's **rhythmicity**.

The heart's attachment system is a **closed oscillation system**. Its tissue and molecular structure is designed to undergo a stretching movement with each beat. It's like a system of **elastic bands** that constantly return the heart to its equilibrium position.

This structuring of the **pericardial fascial tissue** supports the heart, and the heart itself is supported in its rhythmicity by its fascial environment.

The entire fascial system around the heart can influence its rhythm. Hence the importance of the **structural freedom** of the cardiac environment. This is where direct action is possible.

The heart in **systole-diastole** should be seen as an **octopus**, with the **aortic arch** and the entire descending trunk down to the junctions.

It's not just the body, but also its "tentacles" that can be felt. In the belly, in the throat, in the feet, in the fingers.

This implies that a **destabilisation** of the hip can destabilise the heart. There are cases of **arrhythmias** that can be improved by releasing the hips.

There is a complex relationship through the **endothelium**, this tissue that connects the entire system.

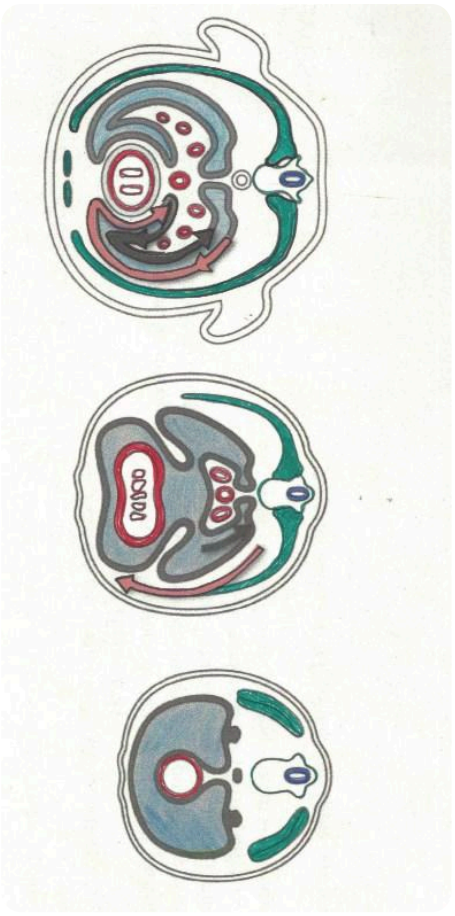


FIG. — Lateral embryonic folding

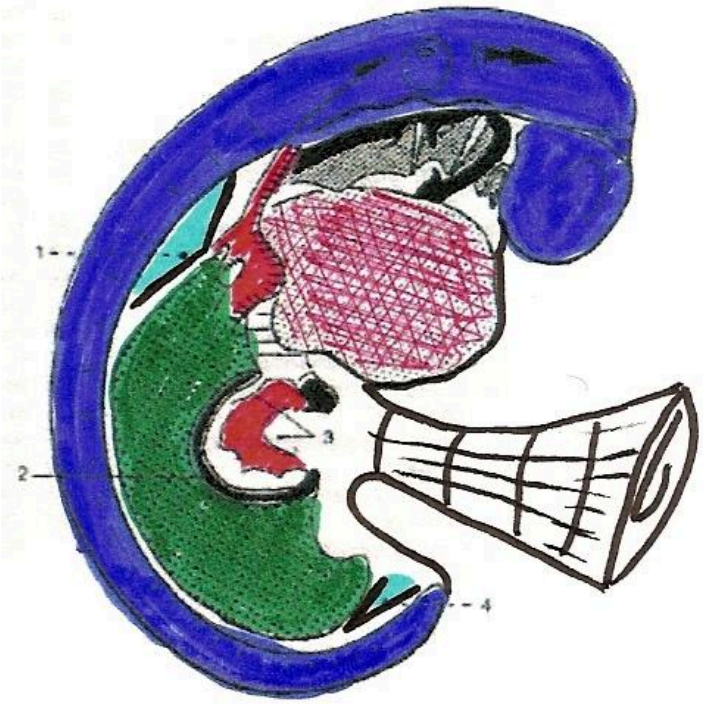


FIG. — Embryo with allantois

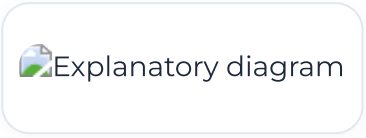


FIG. — Explanatory diagram

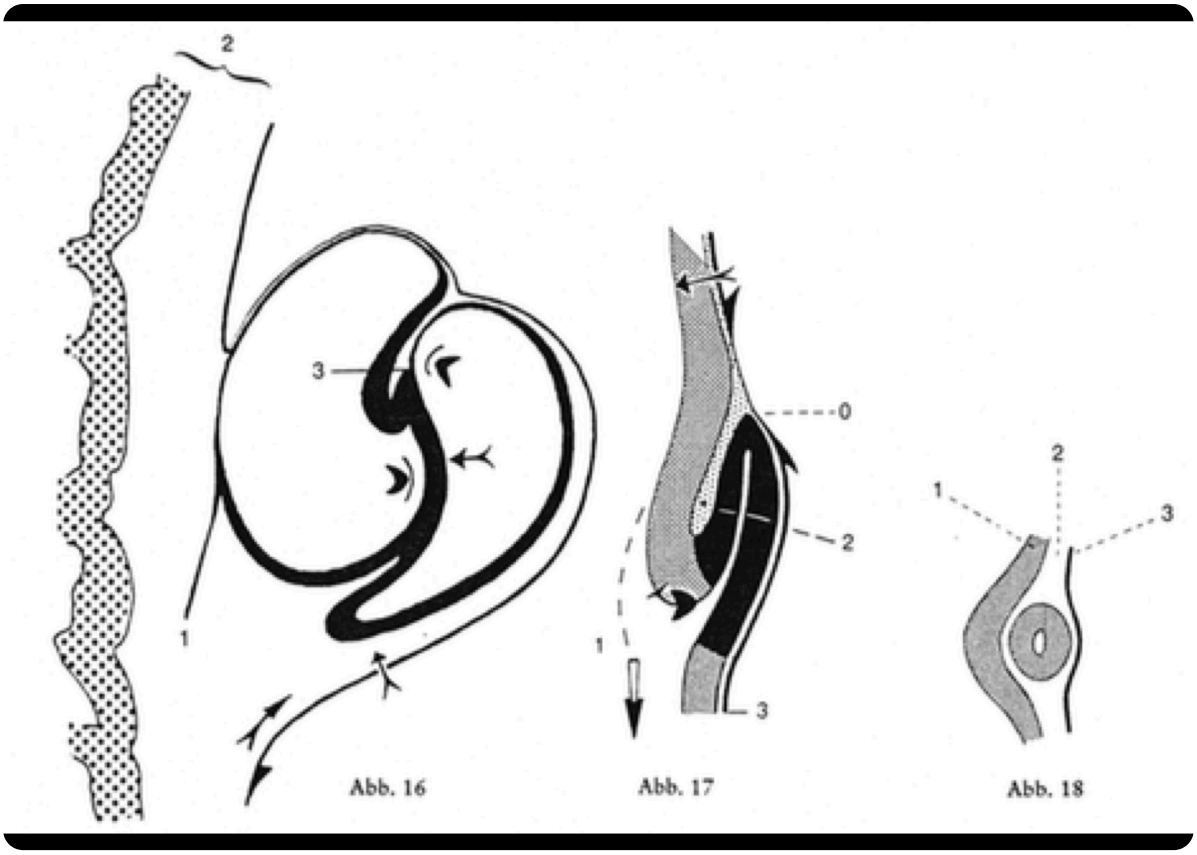


FIG. — Neural tube development

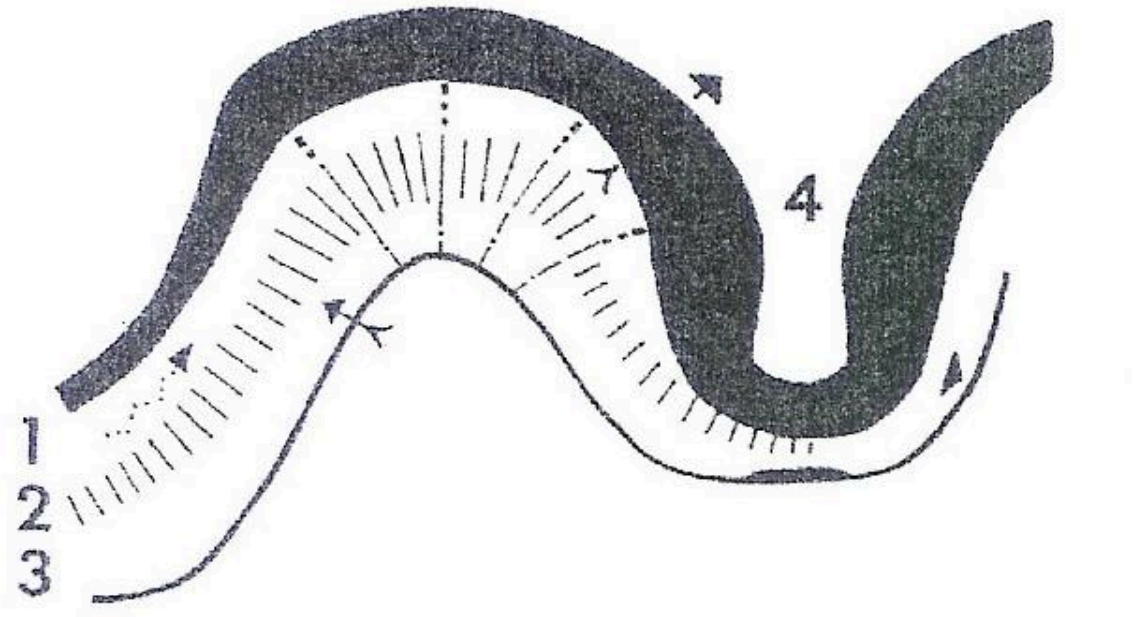
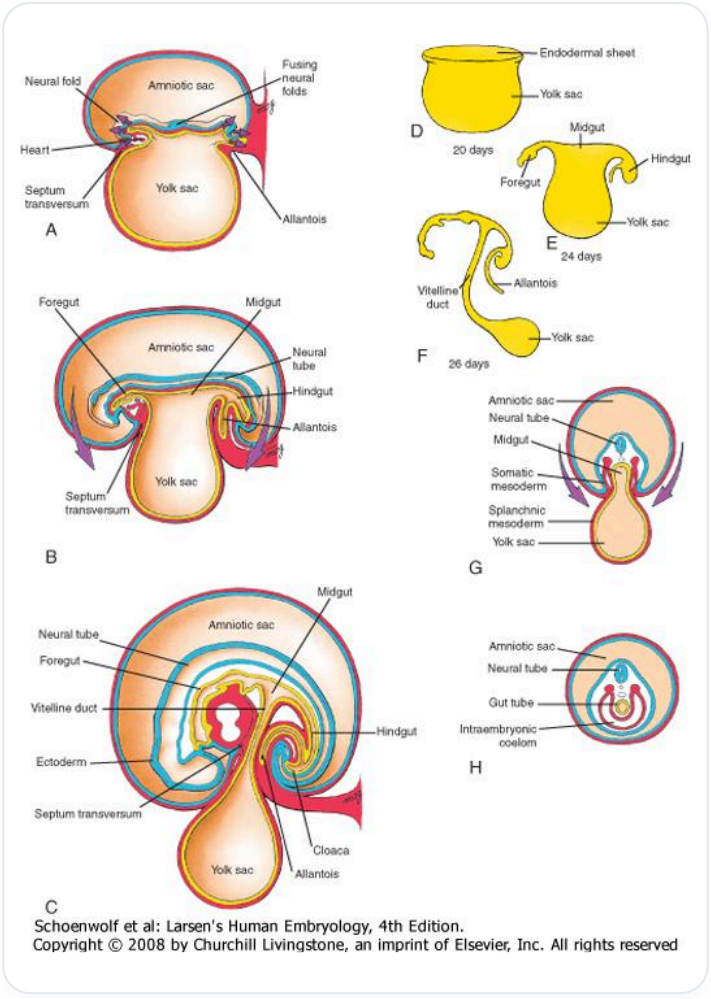


FIG. — Neural plate, groove



Schoenwolf et al: Larsen's Human Embryology, 4th Edition.
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FIG. — Embryonic digestive tract development

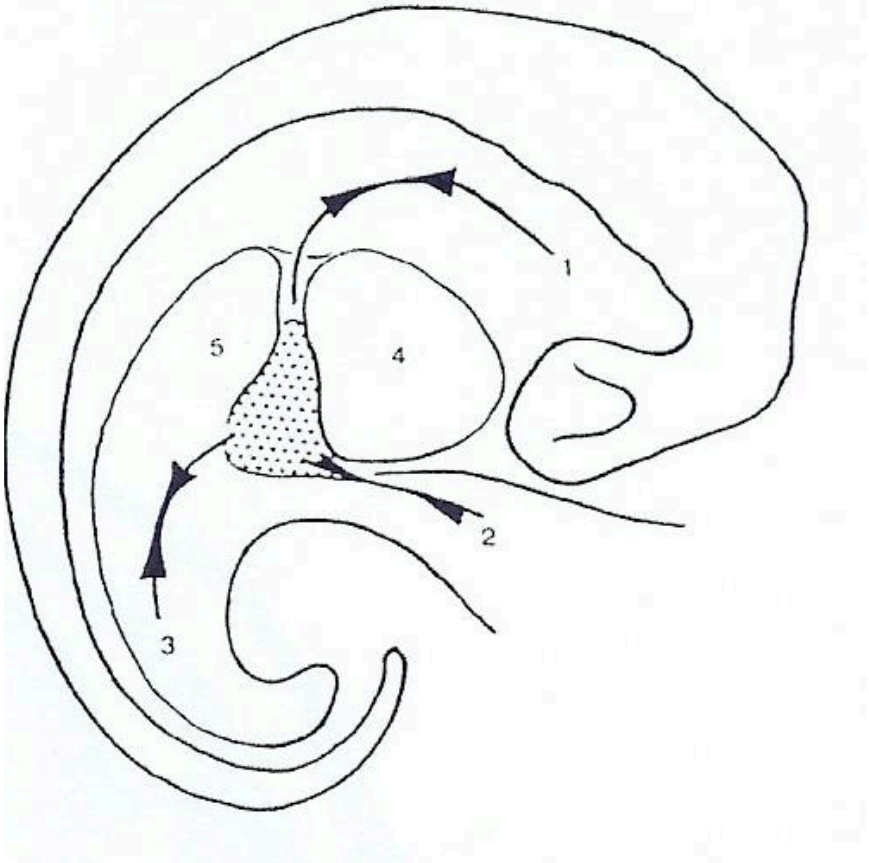


FIG. — Circulation embryo liquide céphalo-rachidien